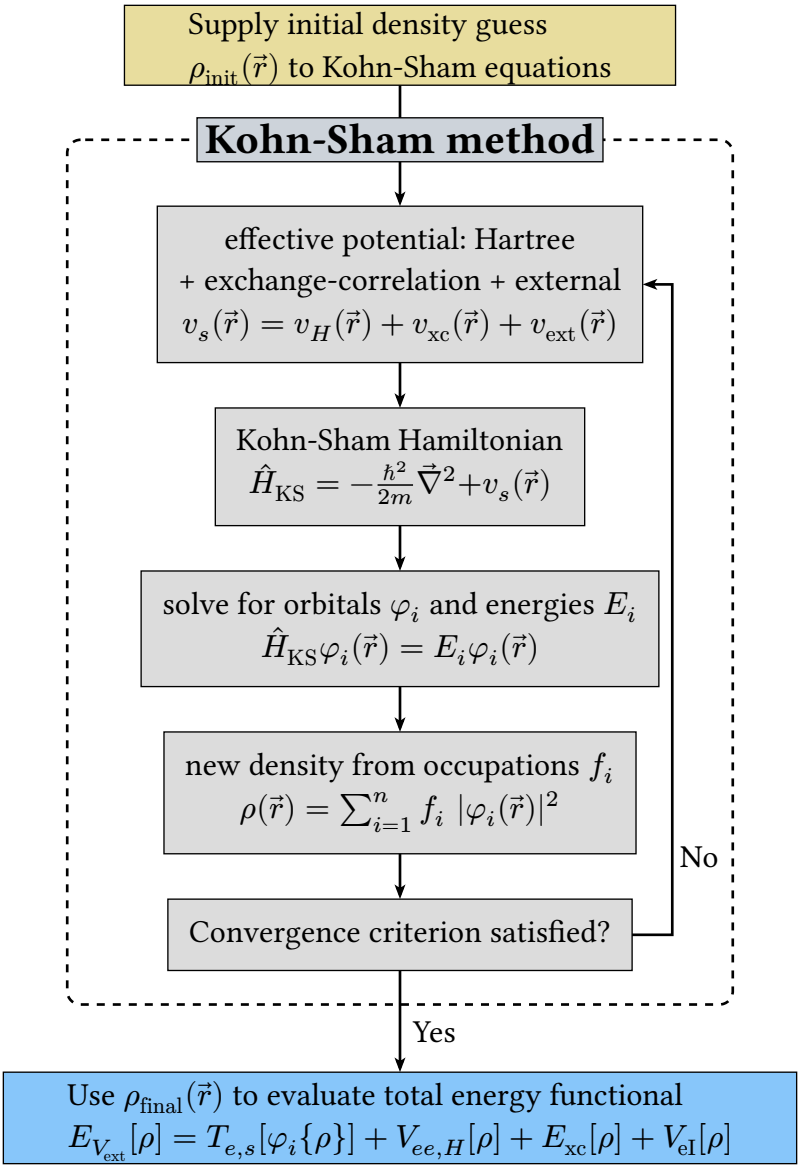
Supply initial density guess
$\rho_{\text{init}}(\vec{r})$ to Kohn-Sham equations
Kohn-Sham method
effective potential: Hartree
+ exchange-correlation + external
$v_s(\vec{r}) = v_H(\vec{r}) + v_{\text{xc}}(\vec{r}) + v_{\text{ext}}(\vec{r})$
Kohn-Sham Hamiltonian
$\hat{H}_{\text{KS}} = -\frac{\hbar^2}{2m}\vec{\nabla}^2 + v_s(\vec{r})$
solve for orbitals φ_i and energies E_i
$\hat{H}_{\text{KS}}\varphi_i(\vec{r}) = E_i\varphi_i(\vec{r})$
new density from occupations f_i
$\rho(\vec{r}) = \sum_{i=1}^{n} f_i \ |\varphi_i(\vec{r})|^2$
Convergence criterion satisfied?
No
Yes
Use $\rho_{\text{final}}(\vec{r})$ to evaluate total energy functional
$E_{V_{\text{ext}}}[\rho] = T_{e,s}[\varphi_i\{\rho\}] + V_{ee,H}[\rho] + E_{\text{xc}}[\rho] + V_{\text{eI}}[\rho]$